

Multimodal Fusion Strategies

Early, late, and intermediate fusion — how machines combine sight, sound, and language into a unified understanding

10 min

Duration

B3 – Apply

Bloom's

L3 – Intermediate

Level

TH-000005 & 000007

Prerequisite

Session Overview

What you will learn and the core challenge of multimodal AI

What You Will Learn

- Why fusion is the core challenge in multimodal AI
- The alignment prerequisite before any fusion can work
- Early, late, and intermediate fusion — definitions and trade-offs
- Cross-modal attention: the transformer mechanism powering modern fusion
- How CLIP, Flamingo, BLIP-2, LLaVA, GPT-4o, and Gemini fuse modalities
- Practical selection guide for AI engineers

Concept at a Glance

- Duration: 10 minutes
- Bloom's Level: B3 – Apply
- Difficulty: L3 – Intermediate
- Module: 4.3 Multimodal Fusion and Training
- Prerequisites: ViT (TH-000002), CLIP (TH-000005), Audio Features (TH-000007)
- Next: Audio-Visual Learning (TH-000011)

Learning Objectives

By the end of this session you will be able to:

10 min

Duration

B3 – Apply

Bloom's

L3 – Intermediate

Level

ViT & CLIP

Prerequisite

- 1 Define multimodal fusion and explain why it is technically challenging
- 2 Compare early, late, and intermediate fusion approaches on their key trade-offs
- 3 Describe cross-modal attention and trace a Visual QA forward pass through it
- 4 Identify which fusion strategy is used in CLIP, GPT-4o, Gemini 1.5, Flamingo, and LLaVA
- 5 Select an appropriate fusion strategy given system constraints and data availability
- 6 Recognise the alignment problem and explain why alignment must precede fusion

SECTION 1 — THE FUSION PROBLEM

Why Multimodal Fusion Is Hard

Each modality has a different data format, temporal structure, and feature space — these must be reconciled

The Core Technical Challenges of Multimodal Fusion

Humans effortlessly combine vision, hearing, and language. Machines must explicitly learn every aspect of this integration

Dimensionality Mismatch

Images produce spatial grids ($H \times W \times C$). Audio is a time series ($T \times F$). Text is a token sequence ($S \times D$). Each modality has a fundamentally different tensor shape — you cannot simply concatenate them without projection.

Temporal Alignment

Video frames at 30 fps and audio at 16,000 samples/sec must be synchronised frame-by-frame. An off-by-100ms misalignment breaks the audio-visual correspondence that models rely on.

Semantic Gap

A pixel value of 128 and a word embedding value of 0.82 mean completely different things. Raw features from different modalities cannot be directly compared — they need to be projected into a shared semantic space.

Modality Dominance

Models over-rely on the dominant modality (text in VLMs) and effectively ignore the weaker one. A VQA model that ignores the image and answers from language priors alone achieves 60%+ accuracy — a known failure mode.

Prerequisite — Alignment: Before any fusion, modalities must be encoded into a common representational space. Image encoder (ViT), audio encoder (wav2vec), text encoder (LLM) — all projected to the same d_{model} dimensionality. Fusion without alignment produces noise.

SECTION 2 — THREE FUSION STRATEGIES

Early, Late, and Intermediate Fusion

Three fundamentally different answers to 'when and how should modalities be combined?'

Early Fusion: Combine Raw Features First

Feature-level fusion: concatenate or combine raw modality representations before any shared deep processing

Definition

Early fusion concatenates or combines raw feature representations from all modalities before passing them to any shared model layers. The unified model learns cross-modal interactions from the very first layer — no modality-specific processing occurs first.

Advantages

- ✓ Captures low-level cross-modal correlations from the first layer
- ✓ Single shared model processes all modalities — fewer parameters
- ✓ Works well when modalities are tightly aligned (synchronised audio-video)
- ✓ Real-time performance: one forward pass through one model

Disadvantages

- ✗ Requires strict temporal and spatial alignment between modalities
- ✗ Missing modality at inference breaks the entire pipeline
- ✗ Highly sensitive to modality-specific noise
- ✗ Difficult to scale to many modalities — grows combinatorially

Example: Early audio-visual models (2016-2018): concatenate visual embedding and audio embedding before a shared classifier. Good for tightly synchronised audio-video event detection. Rarely used in modern VLMs due to missing-modality brittleness.

Late Fusion and Intermediate Fusion

Two more strategies — late fusion for robustness, intermediate fusion for depth of cross-modal learning

Late Fusion (Decision-Level)

Train a separate model for each modality. Combine the final predictions with a lightweight fusion layer (average, vote, small network).

- ✓ Robust to missing modalities at inference
- ✓ Each modality model can be pretrained separately
- ✓ Easy to add/remove modalities
- ✗ No cross-modal learning — misses interactions
- ✗ Duplicate computation for shared semantics
- ✗ Shallow fusion is limited in expressiveness

Example: Ensemble a vision classifier with an audio classifier for audio-visual event detection. Robust, simple, but misses synergies.

Intermediate Fusion ✓ (Modern Standard)

Process each modality through dedicated encoders to extract rich representations. Combine these representations in a shared deep network. Fusion happens at one or more hidden layers.

- ✓ Encoders specialise per modality
- ✓ Shared layers learn deep cross-modal interactions
- ✓ Cross-attention layers attend selectively across modalities
- ✓ Missing modalities can be masked (partially)

Example: GPT-4o, Gemini 1.5, LLaVA, Flamingo, BLIP-2. The LLM backbone acts as the shared fusion network.

Industry consensus: Intermediate fusion is the dominant strategy in 2024–2025 production systems. The LLM backbone serves as the universal fusion network. Modality-specific projectors bridge encoders into the shared token space.

SECTION 3 — CROSS-MODAL ATTENTION

Cross-Modal Attention: The Mechanism

Queries from one modality; Keys and Values from another — how text tokens learn to 'look at' image patches

Cross-Modal Attention Formula

Standard self-attention has Q, K, V from the same sequence. Cross-modal attention routes them across modalities

$$\text{CrossAttn}(Q_{\text{text}}, K_{\text{img}}, V_{\text{img}}) = \text{softmax}(Q_{\text{text}} \cdot K_{\text{img}}^T / \sqrt{d_k}) \cdot V_{\text{img}}$$

Q from text tokens | K, V from image patches | d_k = key dimension for scaling

Q (Query) from text:

The text token 'asks a question': what visual information do I need? The query vector is computed by projecting the text token embedding through W_Q .

K (Key) from image:

Each image patch 'advertises its content': here is what I contain. The key vector is computed by projecting the patch embedding through W_K .

V (Value) from image:

The actual information each patch contributes if selected. The value vector is computed by projecting the patch embedding through W_V .

Attention weights:

$\text{softmax}(Q \cdot K^T / \sqrt{d_k})$ gives a probability distribution over all image patches for each text token. High weight $\rightarrow$ text token focuses on that image patch.

Output:

Weighted sum of V — a context vector for each text token that integrates information from the most relevant image patches. No hard-coded alignment needed.

Visual QA example: Query: 'What colour is the car?' Text tokens compute $Q \rightarrow$ attend to image patches near car regions (high K similarity) $\rightarrow V$ weighted sum tells the model the car's colour. No explicit object detection or spatial annotation needed — the model learns this from data.

SECTION 4 — PRODUCTION FUSION SYSTEMS

CLIP, Flamingo, LLaVA, GPT-4o, Gemini

How leading models actually implement fusion — from contrastive alignment to native end-to-end training

CLIP: Contrastive Cross-Modal Alignment

Radford et al. (OpenAI, 2021) — the foundation of modern multimodal AI: no explicit fusion layer, just shared embedding space

Architecture: Two separate encoders (ViT for images, Transformer for text) trained with a contrastive objective. Matching image-text pairs are pushed together in a shared 512-dimensional embedding space; non-matching pairs are pushed apart. No explicit cross-attention between modalities — fusion happens as a dot product similarity in the shared space.

Why CLIP Works

- 400M image-text pairs from the internet — unprecedented scale
- Zero-shot transfer: classify, retrieve, or compose without task-specific training
- No explicit fusion layer — alignment IS fusion via shared embedding space
- Enables: DALL-E 2, Stable Diffusion conditioning, LLaVA visual encoder

Impact on the Field

- Standard visual encoder in LLaVA, InstructBLIP, BLIP-2, CogVLM
- Foundation for SigLIP (sigmoid loss, Google), OpenCLIP, EVA-CLIP
- Powers most image retrieval and visual search systems in production
- Spawned the entire field of vision-language model research (2021–present)

Key insight: CLIP is not a fusion model in the traditional sense — it's an alignment model. It creates a shared space where you can compare images and text directly. All modern VLMs use CLIP's visual encoder as a frozen backbone.

LLaVA, Flamingo, BLIP-2, and GPT-4o

Four progressively more capable approaches to intermediate fusion — from minimal projectors to native end-to-end

LLaVA (2023) UW + Microsoft

Simplest effective intermediate fusion. CLIP ViT-L/14 (frozen) → 2-layer MLP projector → LLaMA/Vicuna (partially tuned). Image tokens prepended to text tokens. Training cost: ~\$300. Spawned hundreds of variants.

BLIP-2 (2023) Salesforce

Q-Former (Querying Transformer) bridges frozen image encoder and frozen LLM. Only Q-Former is trained — both large models frozen. 32 learned query tokens attend to image. Extremely parameter-efficient. Base for InstructBLIP, MiniGPT-4.

Flamingo (2022) DeepMind

Inserts gated cross-attention layers between frozen LM layers. Perceiver Resampler compresses image patches to fixed-length visual tokens. Gating: $\tanh(\alpha)$ controls visual info flow. Enables few-shot VQA without task-specific fine-tuning.

GPT-4o (2024) OpenAI

Native end-to-end multimodal training across text, image, and audio simultaneously. No separate pipeline. All modalities share Transformer weights. Real-time voice + vision. Enables natural cross-modal reasoning at every layer.

Fusion depth spectrum: LLaVA (shallow MLP bridge) → Flamingo (gated cross-attention) → BLIP-2 (Q-Former bottleneck) → GPT-4o (native joint training). Deeper fusion = richer interaction but more compute and training complexity.

Practical Fusion Strategy Selection Guide

Use this decision framework to choose the right fusion approach for your specific system constraints

Scenario / Constraint	Recommended Strategy	Rationale
Modalities always co-present	Early or intermediate fusion	Can exploit full cross-modal correlation
Modalities sometimes missing	Late fusion or masked cross-attention	Individual modality models work standalone
Limited training data (<10K)	Late fusion (pretrained unimodal models)	Reduces search space; fewer params to train
Building on pretrained LLM	Intermediate (projector — LLaVA style)	LLM as universal fusion backbone
Two large frozen models	BLIP-2 Q-Former pattern	Train only the lightweight bridge
Zero-shot cross-modal retrieval	CLIP-style contrastive alignment	Shared embedding space enables any-to-any search
Real-time, single GPU	Early fusion (single model pass)	Lowest inference latency

Rule of thumb: Intermediate fusion with cross-attention is the production default. Use late fusion when modularity and missing-modality robustness matter more than accuracy. Use CLIP-style alignment when you need retrieval without generation.

Key Takeaways

The six essential points from this concept

Fusion is how AI combines modalities Explicit representation merging of vision, audio, and language into a unified understanding. Not automatic — must be designed and trained.

Alignment must precede fusion Each modality encoder (ViT, wav2vec, LLM) must project to the same `d_model` dimension before fusion can be meaningful. Fusion without alignment produces noise.

Three strategies with distinct trade-offs Early: lowest latency, brittle to missing modalities. Late: most robust, no cross-modal learning. Intermediate: best accuracy, current industry standard.

Cross-modal attention is the mechanism Text tokens query image patches via $Q=\text{text}, K,V=\text{image}$. No hand-coded alignment — the model learns what to attend to from data.

CLIP alignment enables zero-shot transfer Shared embedding space trained with contrastive loss. The foundation of DALL-E 2, LLaVA, and modern VLMs. 400M pairs from web.

LLaVA proves simplicity works A 2-layer MLP projector bridging CLIP ViT and LLaMA achieves GPT-4V-level performance on many benchmarks. ~\$300 to train.

Note: Next: AI-MM-IN-TH-000011 — Audio-Visual Learning (contrastive SSL, AVHuBERT, AudioCLIP, temporal alignment)

AI-MM-IN-TH-000010 Complete

Multimodal Fusion Strategies

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